# Lesson 6 Compare an earthquake hypothesis with other evidence

For slide 28, about six minutes. This exercise summarises published research; it does not report new measurements.

| Evidence | What it contributes / limitation |
|---|---|
| Tree-ring record | Strong growth suppression beginning in 1813 was reported in trees near the fault at Wrightwood. A growth change alone does not identify its cause. |
| Historical record | A damaging earthquake occurred on 8 December 1812. Damage and timing information came from sources other than rings; the location and magnitude remain uncertain. |
| Useful comparisons | Were trees away from the fault also suppressed? Are wounds, tilting or root damage present? Check the original study and its controls. |

**Could an earthquake in winter 1812 produce narrow rings beginning in 1813? Name evidence that helps distinguish earthquake damage from drought. Can ring widths alone establish the day of the earthquake?**

## Check

Damage during dormancy can affect the following growing season. Compare spatial concentration near the fault, damage traces, trees farther away and independent historical accounts. The exact date comes from historical evidence, not ring widths alone.

The SCEDC account also incorporates dendrochronological research. Its historical evidence is the independent component; counting a summary article as an additional independent result would double-count evidence.

Jacoby, Sheppard & Sieh (1988), Science 241:196–199

SCEDC: 1812 Wrightwood earthquake